

Task ECS

1. Setup a highly available ECS cluster with load balancer and dynamic port mapping.

NOTE:

Your cluster should maintain at least 6 tasks at any point of time and should be highly available across multiple AZ's.

Use the below image from deployment:

sabair0509/hiring-app:works

➤ Create a cluster app

Create cluster [Info](#)

An Amazon ECS cluster groups together tasks and services, and allows for shared capacity and common configurations. All of your tasks, services and capacity must belong to a cluster.

Cluster configuration

Cluster name

Cluster name must be 1 to 255 characters. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

► Service Connect defaults – optional

▼ Infrastructure - advanced [Info](#)

Configure the manner of obtaining compute resources that will be used to host your application.

Select a method of obtaining compute capacity

Your cluster is automatically configured for AWS Fargate (serverless), but you may choose to add Amazon EC2 instances (servers).

☐ Fargate only
Serverless – you don't think about creating or managing servers. Great for most common workloads.

☐ Fargate and managed instances
Managed instances - Amazon ECS will manage patching and scaling on your behalf while giving you configurability about the types of instances. Great for more advanced workloads.

☒ Fargate and Self-managed instances
Self-managed instances - you must ensure the instances are patched and scaled properly, and you have full control over the instances.

➤ Auto scaling and on demand select

Auto Scaling group (ASG) [Info](#)

☒ Create a new Auto Scaling group - advanced

☐ Use an existing Auto Scaling group

Provisioning model

Select a provisioning model for your instances

☒ On-demand
With on-demand instances, you pay for compute capacity by the hour, with no long-term commitments or upfront payments.

☐ Spot
Amazon EC2 Spot instances let you take advantage of unused EC2 capacity in the AWS cloud. [Find out more](#)

Container instance Amazon Machine Image (AMI)

Choose the Amazon ECS-optimised AMI for your instance.

EC2 instance type

Choose based on the workloads you plan to run on this cluster.

x86_64
2 vCPU 4 GiB memory

EC2 instance role
An instance role is used by Amazon EC2 instances to make AWS API requests. If you don't already have an instance IAM role created, we can create one for you.

Create default role ▼  [Create new instance profile](#)

Desired capacity
Specify the number of instances to launch in your cluster.

Minimum **Maximum**

SSH Key pair
If you do not specify a key pair, you can't connect to the instances via SSH unless you choose an AMI that is configured to allow users another way to log in.

aws ▼  [Create a new key pair ↗](#)

Root EBS volume size
You can increase the size of the root EBS volume to allow for greater image and container storage.

A min of 30 GiB and a max of 16,384 GiB is allowed.

➤ Networking Settings default in different VPC subnets

▼ Network settings
The default VPC, subnets and security groups are selected by default.

VPC
Select a VPC to use for your Amazon ECS resources.

vpc-096ec8a65d98c6729
default ▼  [Create a new VPC ↗](#)

Subnets
Select the subnets where your instances are launched and your tasks run. We recommend that you use three subnets for production.

Choose subnets ▼ [Clear current selection](#)

subnet-05892b6b6d9730f66 ✕
pu1
us-east-1a 172.31.0.0/20

subnet-0a25ef5df35935262 ✕
us-east-1c 172.31.16.0/20

Security group | [Info](#)
Choose an existing security group or create a new security group.

☒ Use an existing security group
☐ Create a new security group

Security group name
Choose an existing security group.

Choose security groups ▼

sg-04eeb95c40a6368d8 ✕
default

➤ Cluster created

[vice](#) > Clusters

 Cluster Hiring-App-Cluster1 has been created successfully.

Clusters (1) [Info](#)

By default, we only load up to 1,000 clusters at a time.

Cluster	Services	Tasks	Container instances	CloudWatch monitor
Hiring-App-Cluster1	0	No tasks running	2 EC2	 Default

➤ Two EC2 Instances Created Automatically

Instances (2) Info Last updated less than a minute ago Connect Instance state ▼

Saved filter sets
Running ▼

Instance state = running X Clear filters ▼

☐	Name	Instance ID	Instance state	Instance type	Status check	Availab
☐	ECS Instance - ...	i-015b007ffb94c7f8c	Running	c7i-flex.large	Initializing	us-east-
☐	ECS Instance - ...	i-01e5c89fb9e556409	Running	c7i-flex.large	Initializing	us-east-

➤ Role Created

IAM > Roles > Create role

Step 1: Select trusted entity (selected)
Step 2: Add permissions
Step 3: Name, review, and create

Select trusted entity Info

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case

Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
Elastic Container Service ▼

Choose a use case for the specified service.

Use case

- ☐ Elastic Container Service
Allows ECS to create and manage AWS resources on your behalf.
- ☐ Elastic Container Service Autoscale
Allows Auto Scaling to access and update ECS services.
- ☒ **Elastic Container Service Task**
Allows ECS tasks to call AWS services on your behalf.
- ☐ EC2 Role for Elastic Container Service
Allows EC2 instances in an ECS cluster to access ECS.
- ☐ Elastic Container Service for VPC Lattice
Allows access to create and manage AWS service resources required to manage VPC Lattice feature in ECS workloads

- Step 2
● Add permissions
- Step 3
● **Name, review, and create**

Role details

Role name

Enter a meaningful name to identify this role.

ecs-task-execution

Maximum 64 characters. Use alphanumeric and '+', '-', '@', '_' characters.

Description

Add a short explanation for this role.

Allows ECS tasks to call AWS services on your behalf.

Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: '_', '+', '@', '-', '/', '[', ']', '!', '\$', '%', '&', '*', '=', ':', ';', ',', '<', '>', '"', ''', '`', '|', '>', '<', '>', '<', '>', '<'.

Step 1: Select trusted entities

Trust policy

```
1 {  
2   "Version": "2012-10-17",
```

- Step 1
● Select trusted entity
- Step 2
● **Add permissions**
- Step 3
○ Name, review, and create

Add permissions [Info](#)

Permissions policies (1/1163) [Info](#)

Choose one or more policies to attach to your new role.

Q AmazonECSTask

Filter by Type

All types

1 match

☒ Policy name [↗](#)

▲ Type ▼

☒ [AmazonECSTaskExecutionRolePolicy](#)

AWS managed

► Set permissions boundary - *optional*

➤ Created New Task Definition

Amazon Elastic Container Service

Express Mode
Clusters
Namespaces
Task definitions
Daemon task definitions
Account settings

[AWS Resource Explorer](#) [↗](#)
[AWS Batch](#) [↗](#)
[Amazon ECR](#) [↗](#)
[Repositories](#) [↗](#)

[Online learning workshop](#) [↗](#)
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Create new task definition [Info](#)

Task definition configuration

Task definition family [Info](#)

Specify a unique task definition family name.

hiring-app-definition

Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.

▼ Infrastructure requirements [Info](#)

Specify the infrastructure requirements for the task definition.

Launch type [Info](#)

Selection of the launch type will change task definition parameters.

☐ AWS Fargate

Serverless compute for containers.

☐ Managed instances

Use if you have specific hardware constraints, such as GPU accelerators, CPU instruction sets, or network-optimized hardware (offloads scaling, patching, and instance management to AWS).

☒ Amazon EC2 instances

Self-managed infrastructure using Amazon EC2 instances.

OS, Architecture, Network mode

Network mode is used for tasks and is dependent on the compute type selected.

Operating system/Architecture

Info

Linux/X86_64

Network mode

Info

bridge

Task size

Info

Specify the amount of CPU and memory to reserve for your task.

CPU

1 vCPU

Memory

1 GB

▼ Task roles - conditional

Task role

Info

A task IAM role allows containers in the task to make API requests to AWS services.

ecs-task-execution

↻

Create new role

Task execution role

Info

A task execution IAM role is used by the container agent to make AWS API requests on your behalf. If you don't already have a task execution IAM

ecs-task-execution

↻

Create new role

➤ Select New Name and Image URL

Specify a name, container image and whether the container should be marked as essential. Each task definition must have at least one essential container.

Name

hiring-app

Up to 255 letters (uppercase and lowercase), numbers, hyphens, and underscores are allowed.

Essential container

Yes

Image URI

sabair0509/hiring-app:works

Browse ECR images

Up to 255 letters (uppercase and lowercase), numbers, hyphens, underscores, colons, periods, forward slashes, and number signs are allowed.

Private registry

Info

Store credentials in Secrets Manager, and then use the credentials to reference images in private registries.

☒ Private registry authentication

Port mappings

Info

Add port mappings to allow the container to access ports on the host to send or receive traffic. For port name, a default will be assigned if left blank.

Host port	Container port	Protocol	Port name	App protocol	
0	8080	TCP	container-port-prot	HTTP	Remove

Add port mapping

➤ Task created

Elastic Container Service > Task definitions > hiring-app-definition > Revision 1 > Containers

lastic Service

ns

definitions

ngs

Explorer

Task definition successfully created

hiring-app-definition:1 has been successfully created. You can use this task definition to deploy a service or run a task.

View task definit

hiring-app-definition:1

Last updated 8 June 2026, 15:02 (UTC+5:30)

Deploy

Actions

Create new

Overview

Info

ARN	arn:aws:ecs:us-east-1:9284300965:03:task-definition/hiring-app-definition:1	Status	ACTIVE	Time created	8 June 2026, 15:02 (UTC+5:30)	App environment	EC2
Task role	ecs-task-execution	Task execution role	ecs-task-execution	Operating system/Architecture	Linux/X86_64	Network mode	bridge
Fault injection	Turned off						

➤ Create Target Group

Create target group

A target group can be made up of one or more targets. Your load balancer routes requests to the targets in a target group.

Settings - *immutable*

Choose a target type and the load balancer and listener will route traffic to your target. These settings can't be modified.

Target type

Indicate what resource type you want to target. Only the selected resource type can be registered to this target group.



Instances

Supports load balancing to instances in a VPC. Integrate with Auto Scaling Groups or ECS services for automatic management.

Suitable for: ALB NLB GWLB



IP addresses

Supports load balancing to VPC and on-premises resources. Facilitates routing to IP addresses and network interfaces on the same instance. Supports IPv6 targets.

Suitable for: ALB NLB GWLB



Lambda function

Supports load balancing to a single Lambda function. ALB required as traffic source.

Suitable for: ALB



Application Load Balancer

Allows use of static IP addresses and PrivateLink with an Application Load Balancer. NLB required as traffic source.

Suitable for: NLB

Target group name

Name must be unique per Region per AWS account.

hiring-tg

VPC

Select the VPC with the instances that you want to include in the target group. Only VPCs that support the IP address type selected above are available in this list.

vpc-096ec8a65d98c6729
172.31.0.0/16

(default) ▼



Create

Protocol version

☒ HTTP1

Send requests to targets using HTTP/1.1. Supported when the request protocol is HTTP/1.1 or HTTP/2.

☐ HTTP2

Send requests to targets using HTTP/2. Supported when the request protocol is HTTP/2 or gRPC, but gRPC-specific features are not available.

☐ gRPC

Send requests to targets using gRPC. Supported when the request protocol is gRPC.

Health checks

The associated load balancer periodically sends requests, per the settings below, to the registered targets to test their status.

Health check protocol

HTTP ▼

Health check path

Use the default path of "/" to perform health checks on the root, or specify a custom path if preferred.

/

Up to 1024 characters allowed

2 selections are now pending below. Include more or register targets when ready.

Review targets

Targets (2) Remove all pending

Filter targets ☐ Show only pending < 1 > ⚙️

Instance ID	Name	Port	State	Security groups	Zone	Private IPv4 address
i-015b007ffb94c7f8c	ECS Instance - Hiring-App-Cluster1	80	Running	default	us-east-1a	172.31.14.195
i-01e5c89fb9e556409	ECS Instance - Hiring-App-Cluster1	80	Running	default	us-east-1c	172.31.24.57

2 pending Cancel Previous

➤ Target Group created

EC2 > Target groups > hiring-tg

EC2

- Dashboard
- AWS Global View
- Events
- Instances**
 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- Images

hiring-tg

Successfully created the target group: **hiring-tg**. Anomaly detection is automatically applied to all registered targets. Results can be viewed in the Targets tab.

Details

arn:aws:elasticloadbalancing:us-east-1:928430096503:targetgroup/hiring-tg/4bdc048e5435fd20

Target type Instance	Protocol : Port HTTP: 80	Protocol version HTTP1	VPC vpc-096...
IP address type IPv4	Load balancer None associated		

2 Total targets	0 Healthy	0 Unhealthy	2 Unused	0 Initial
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0 Anomalous

➤ Create Application Load Balancer created

Create Application Load Balancer [Info](#)

The Application Load Balancer distributes incoming HTTP and HTTPS traffic across multiple targets such as Amazon EC2 instances. When a connection request, it evaluates the listener rules in priority order to determine which rule to apply, and if applicable, it selects a target to forward the request to.

► How Application Load Balancers work

Basic configuration

Load balancer name
Name must be unique within your AWS account and can't be changed after the load balancer is created.

A maximum of 32 alphanumeric characters including hyphens are allowed, but the name must not begin or end with a hyphen.

Scheme [Info](#)
Scheme can't be changed after the load balancer is created.

☒ **Internet-facing**

- Serves internet-facing traffic.
- Has public IP addresses.
- DNS name resolves to public IPs.
- Requires a public subnet.

☐ **Internal**

- Serves internal traffic.
- Has private IP addresses.
- DNS name resolves to private IPs.
- Compatible with the IPv4 and Dualstack IP address types.

Load balancer IP address type [Info](#)
Select the front-end IP address type to assign to the load balancer. The VPC and subnets mapped to this load balancer must include the selected IP address type.

☒ **IPv4**
Includes only IPv4 addresses.

Network mapping Info

The load balancer routes traffic to targets in the selected subnets, and in accordance with your IP address s

VPC Info

The load balancer will exist and scale within the selected VPC. The selected VPC is also where the load balancer targets mus premises targets, or if using VPC peering. To confirm the VPC for your targets, view [target groups](#).

vpc-096ec8a65d98c6729
172.31.0.0/16

IP pools Info

You can optionally choose to configure an IPAM pool as the preferred source for your load balancers IP addresses. Create or

☐ Use IPAM pool for public IPv4 addresses
The IPAM pool you choose will be the preferred source of public IPv4 addresses. If the pool is depleted IPv4 addresses v

Availability Zones and subnets Info

Select at least two Availability Zones and a subnet for each zone. A load balancer node will be placed in each selected zone

☒ us-east-1a (use1-az1)

Subnet
Only CIDR blocks corresponding to the load balancer IP address type are used. At least 8 available IP addresses are requ internet gateway (IGW) routes configured to allow traffic.

subnet-05892b6b6d9730f66 (pu1)
172.31.0.0/20
Network ACL ALLOW rules: Inbound (1) Outbound (1) | Route table: IGW (✓)

☒ us-east-1c (use1-az4)

Subnet

Security groups Info

A security group is a set of firewall rules that control the traffic to your load balancer. Listed security groups are in the same VPC as you selected above. Selected security groups mu outbound rule to allow traffic to your load balancer.

Security groups
Select up to 5 security groups
default (sg-04eeb95c40a6368d8) X
Inbound rule (14) Outbound rule (2)
Create security g

Default action Info

The default action is used if no other rules apply. Choose the default action for traffic on this listener.

Routing action
☒ Forward to target groups ☐ Redirect to URL ☐ Return fixed re

Forward to target group Info
Choose a target group and specify routing weight or [create target group](#).

Target group	Weight	Percent
hiring-tg Target type: Instance, IPv4 Target stickiness: Off	HTTP 1 0-999	100%

+ Add target group
You can add up to 4 more target groups.

➤ Created ALP

EC2 > Load balancers > hiring-app-lb

C2

Dashboard

AWS Global View

Events

Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

Pages

Successfully created load balancer: **hiring-app-lb**

It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.

hiring-app-lb

▼ Details

Load balancer type Application	Status Provisioning	VPC vpc-096ec8a65d98c6729	Load balancer IP address type IPv4
Scheme Internet-facing	Hosted zone Z355XDOTRQ7X7K	Availability Zones subnet-05892b6b6d9730f66 us-east-1a (use1-az1) subnet-0a25ef5df35935262 us-east-1c (use1-az4)	Date created June 8, 2026, 15:05 (UTC+05:30)
Load balancer ARN arn:aws:elasticloadbalancing:us-east-1:928430096503:loadbalancer/app/hiring-app-lb/52ae68ed250bb26f		DNS name hiring-app-lb-676251875.us-east-1.elb.amazonaws.com (A Record)	

➤ Create service in Cluster App

Notifications 0 0 2 0 0

Create service [Info](#)

Service details

Task definition family
Select an existing task definition. To create a new task definition, go to [Task definitions](#).

hiring-app-definition

Task definition revision Latest
Select the task definition revision from the 100 most recent entries, or enter a revision. Leave the field blank to use the latest revision.

1

Service name
Assign a service name that is unique for this cluster.

hiring-app-definition-service

Up to 255 letters (uppercase and lowercase), numbers, underscores and hyphens are allowed. Service names must be unique within a cluster.

▼ Compute configuration - *advanced*

Compute options [Info](#)

To ensure task distribution across your compute types, use appropriate compute options.

☐ Capacity provider strategy

Specify a launch strategy to distribute your tasks across one or more capacity providers.

☒ Launch type

Launch tasks directly without the use of a capacity provider strategy.

Launch type [Info](#)

Select either managed capacity (Fargate), or custom capacity (EC2 or user-managed, External instances). External instances are registered to your cluster using the EC

EC2

► Troubleshooting configuration - *recommended*

➤ Desired 6

Deployment configuration

Scheduling strategy [Info](#)

☒ **Replica**

Place and maintain a desired number of tasks across your cluster.

☐ **Daemon**

Place and maintain one copy of your task on each container instance.

Desired tasks

Specify the number of tasks to launch.

6

Availability Zone re-balancing [Info](#)

☒ **Turn on Availability Zone re-balancing**

Amazon ECS automatically detects Availability Zone imbalances in task distributions across an ECS service, and evenly redistributes ECS service

Health check grace period [Info](#)

0

seconds

➤ Select Load Balancer App

▼ Load balancing - *optional*

Configure load balancing using Amazon Elastic Load Balancing to distribute traffic evenly across the healthy tasks in your service.

☒ **Use load balancing**

VPC

Select the VPC for your load balancing resources.

vpc-096ec8a65d98c6729
default



[Create a new VPC](#)

Load balancer type [Info](#)

Specify the load balancer type to distribute incoming traffic across the tasks running in your service.

☒ **Application Load Balancer**

An Application Load Balancer makes routing decisions at the application layer (HTTP/HTTPS), supports path-based routing, and can route requests to one or more ports.

☐ **Network Load Balancer**

A Network Load Balancer makes routing decisions at the transport layer (TCP/UDP).

Container

The container and port to load balance the incoming traffic to

hiring-app 8080:8080

Host port:Container port

Container

The container and port to load balance the incoming traffic to

hiring-app 8080:8080

Host port:Container port

Application Load Balancer

Specify whether to create a new load balancer or choose an existing one.

Create a new load balancer

Use an existing load balancer

Load balancer

Choose an existing load balancer to distribute traffic. View existing load balancers and create new one in [EC2 Console](#).

hiring-app-lbinternet-facing

hiring-app-lb-676251875.us-east-1.elb.amazonaws.com

Listener

Specify the port and protocol that the load balancer will listen for connection requests on.

Create new listener

Use an existing listener

Listener

HTTP:80

Listener

Specify the port and protocol that the load balancer will listen for connection requests on.

Create new listener

Use an existing listener

Listener

HTTP:80

Listener rules for 80:HTTP (1)

Traffic received by the listener is routed according to its rules. Rules are evaluated in priority order, from the lowest value to the highest value. The default rule is evaluated last.

< 1 >

Priority	Rule path	Target group
default	/	hiring-tg

Target group

Specify whether to create a new target group or choose an existing one that the load balancer will use to route requests to the tasks in your service.

Create new target group

Use an existing target group

Target group name

hiring-tg

Health check path

/

Health check protocol

HTTP

➤ Service created

Services (1) Info									
8 June 2026, 15:08 (UTC+5:30) Manage tags Update Delete service Create									
Filter services by value Filter launch typeAny launch type Filter scheduling strategyAny scheduling strategy Filter resource management typeAny resource management type < 1 >									
☐	Service name	▲	ARN	Status	Schedu...	Lau...	Task de...	Deployments and tasks	
☐	hiring-app-definition-service		arn:aws:ecs:us-e	Active	Replica	EC2	hiring-ap...		0/6 tas

➤ Task Running Successful

Tasks (1/5) 8 June 2021

Filter tasks by property or value

Filter desired status: Any desired status

Filter launch type: Any launch type

	Task	Last status	Desired status	Task definition	Health status
☒	 02186c3ebfa6485bbd559...	 Running	 Running	hiring-app-definition:1	 Unkno
☐	 2bc7f0f73d9b483fb5781...	 Running	 Running	hiring-app-definition:1	 Unkno
☐	 533720ceee9e4a08924f1...	 Running	 Running	hiring-app-definition:1	 Unkno

➤ Finally check with ALP DNS URL

